

What Is an LLM Judge Worth?

Human Labels Saved Beyond the Best Annotation Design

Anonymous ACL submission

Abstract

LLM judges are increasingly credited with the human annotation they save. That credit is usually measured against uniform annotation: label a random subset, then correct the judge’s bias with the labels. We measure it instead against the cheapest annotation design that uses no judge, for a concrete decision: certifying that a selected system is within ε of the best of a menu. We define a judge’s value as the fraction of human labels it saves for that decision (HES), and measure it in three domains: retrieval (nine fully judged collections), machine translation (WMT22 MQM) and open-ended chat (LMArena), with eleven judges, pre-registered hypotheses and a held-out locked test. [Abstract numbers: (i) decision-aware human designs save X–Y%; judges add A–B% on top; (ii) inflation; (iii) pair-conditional value, accuracy vs ρ ; (iv) pilot predicts relative cost, not near-threshold adoption.]

1 Introduction

Human evaluation is the reference against which NLP systems are compared, and it is expensive. LLM judges promise to stretch it: label everything with the judge, and spend human effort only on correcting the judge’s bias (Angelopoulos et al., 2023a,b; Boyeau et al., 2024; Chen et al., 2026). A growing line of work reports how much this saves, as an effective sample size or a ratio of human labels (Boyeau et al., 2024; Gao et al., 2026; Dorner et al., 2025).

These savings are almost always computed against a single baseline: humans label a uniformly random subset of the evaluation items. But uniform sampling is rarely the best use of a human budget. When the question is a decision, *is system A within ε of the best alternative?*, many items cannot move the answer (two systems return the same output, or a document lies outside both systems’ cutoffs), and others move it a lot. An annotation design

that knows this, with no judge at all, can need far fewer labels. A judge’s saving over uniform sampling therefore mixes two things: what the judge contributes, and what any decision-aware design would have contributed anyway.

We separate the two. For a decision d (a menu of systems, a tolerance ε , a confidence level), an annotation design s and a judge f , we measure $J_\tau(s, f; d)$, the number of human labels, pilot included, at which the decision is certified with probability τ , and define the judge’s *human-evidence saving* $HES_s(f; d) = 1 - J_\tau(s, f; d)/J_\tau(s, \emptyset; d)$. We report HES over the best judge-free design rather than over uniform sampling, and the difference between the two, which we call *inflation*.

Contributions.

1. **Most of the saving belongs to the design.** [Retrieval: decision-weight sampling saves 41–57% of the human labels of uniform sampling; the best judge saves a further 3–20%. MT: ... Chat: ...]
2. **Judge value is inflated when measured over uniform sampling.** [MT/IR numbers.]
3. **Judge value is decision-conditional.** The same judge saves different amounts for different system pairs; across judges and pairs, the saving follows the correlation ρ between true and judge-predicted paired differences, not label accuracy [(numbers per domain)].
4. **A small pilot predicts relative cost, not adoption near the threshold.** [...] In a locked test on a retrieval collection never used in development the pilot ranked designs correctly but over-estimated the judge’s own saving.

We introduce no estimator: the corrected estimator is PPI++ (Angelopoulos et al., 2023b) and the allocation is active statistical inference (Zrnic and

Candès, 2024) for the functional the decision requires. The contribution is the measurement target (labels to a decision, against the best judge-free design), its evaluation across three NLP evaluation settings under pre-registration, and what it shows.

2 Setup

Decision. A menu $\mathcal{M}$ of systems is evaluated on a fixed set of N items with human utilities $u_m(i)$; μ_m is the mean over the set. An audit selects a candidate $\hat{m}$ on a pilot and *certifies* it if, for every competitor j , a one-sided upper confidence bound on $\mu_j - \mu_{\hat{m}}$ is at most ε (Bonferroni over competitors, $\alpha = 0.10$). A certificate is *wrong* if $\max_j \mu_j - \mu_{\hat{m}} > \varepsilon$.

Designs. After a fully labelled pilot of P items, human labels are drawn by Poisson sampling with inclusion probabilities π_i ; the paired difference $D_j = u_j - u_{\hat{m}}$ is estimated by

$$\hat{\Delta}_j = \frac{1}{N} \left[\sum_{i \in \text{pilot}} D_j(i) + \sum_{i \notin \text{pilot}} \lambda_j \hat{D}_j(i) + \sum_{i \in S} \frac{D_j(i) - \lambda_j \hat{D}_j(i)}{\pi_i} \right] \quad (1)$$

where $\hat{D}_j$ is the judge’s predicted difference and $\lambda_j \geq 0$ is fitted on the pilot and then frozen ($\lambda = 0$: humans only). The estimator is unbiased for any judge given the pilot; its variance is estimated by $\sum_S (1 - \pi_i) R_i^2 / \pi_i^2 / N^2$ with $R = D - \lambda \hat{D}$. *Uniform* designs use constant π and label every system’s output of a sampled item. *Decision-aware* designs use the structure of the decision: they never label an item whose difference is known to be zero (identical outputs; documents outside every cutoff), label a shared output once for all comparisons, and sample the rest in proportion to a label-free proxy of $|D|$ (closed-form decision weights in retrieval, output dissimilarity in MT). *Active* designs replace the proxy by the pilot-estimated residual variance of the judge (Zrnic and Candès, 2024).

Cost and metric. Cost counts unique human labels, pilot included. Each budget is a separate pre-committed audit; J_τ interpolates the certification rate over budgets ($\tau = 0.5$ primary, 0.8 where reached). HES, inflation and selection regret are defined in the introduction and Section 7; intervals are paired bootstraps over audits.

3 Domains, judges and protocol

Retrieval. Policies are three cutoff rules over a shared ranked pool [(describe)]; four development

collections (ANTIQUE, CAsT 2019, DBpedia-Entity, TREC DL 21–23), two collections used for a locked rule test (TREC-COVID, Touché) and one never used in development (NeuCLIRBench). Judges: Qwen3-8B (UMBRELA prompt), Qwen3-Reranker-0.6B, the inverted reranker (adversarial).

Machine translation. WMT22 MQM (Freitag et al., 2021) en→de (1,315 segments) and zh→en (1,875), the top four WMT submissions by human MQM. Judges: COMET-22 (Rei et al., 2022), Qwen3-8B and Mistral-7B with the GEMBA-DA prompt (Kocmi and Federmann, 2023), chrF (development), inverted COMET (adversarial).

Chat. LMArena human-preference battles (Chiang et al., 2024), the six model pairs with the most decisive English single-turn battles (494–922 per pair). Judges: Qwen3-8B and Mistral-7B as pairwise judges (both orders averaged), “the longer response wins”, and inverted Qwen3-8B.

Pre-registration. Hypotheses H1–H5 for MT and chat were committed before any LLM or COMET judgment was joined with a human label (Appendix [X]); the retrieval held-out test followed an earlier lock.

4 Where the savings come from

[Figure 1 waterfall; Table 1 J50 per domain; H1.]

5 Judge value is inflated over uniform sampling

[Figure 2; H2.]

6 Judge value is decision-conditional

[Figure 3 heatmap + accuracy vs rho; H3. Retrieval query-level evidence; MT pairs; Arena pairs.]

7 Choosing a judge from a pilot

[Table 2 selection: accuracy, rho, pilot-cost, global-best, humans-only, oracle; H4, H5; NeuCLIR locked result.]

8 Related work

Judges as system rankers. JuStRank (Gera et al., 2025) and Gao et al. (2025) show that instance-level judge accuracy does not determine system-level ranking quality; Zhang et al. (2026) show that judgment quality and downstream utility need not align. We move the target one step further, to the human labels a decision requires.

Prediction-powered evaluation. PPI and PPI++ (Angelopoulos et al., 2023a,b), active inference (Zrnic and Candès, 2024), AutoEval (Boyeau et al., 2024) and prediction-powered ranking (Chatzi et al., 2024) on Chatbot Arena, efficient noisy-judge inference (Chen et al., 2026), PRECISE for ranking metrics (Divekar, 2026), and PPSR, which ranks MT metrics by annotation saved (Gao et al., 2026). All of these measure savings over uniform labelling; Dorner et al. (2025) bound them by a factor of two. We measure them over the best judge-free design.

Annotation design for evaluation. Active comparison of models (Sawade et al., 2012), active testing (Kossen et al., 2021), topic-set design and active judgment selection in IR (Sakai, 2016; Li and Kanoulas, 2017).

9 Limitations

[fill]

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A Pre-registration record

[lock text, deviations]